

Advanced Python

Day 8

Class Variables vs Instance Variables:
Understanding Shared Data vs Personal Data inside Objects

DAILY SESSION OBJECTIVES



Time Focus

2 Hours dedicated to deep-diving into object data structures and memory management.



Primary Goal

Master the distinction between **shared** global data and **personal** local object data.



Real Modeling

Apply these concepts to real-world systems like Schools, Companies, and Game mechanics.

| THE SCHOOL ANALOGY

Types of Information

Personal Data (Unique):

- Student Name (Saurav vs Rahul)
- Roll Number & Grade Marks

Shared Data (Common):

- School Name (ABC School)
- Uniform Color & Principal's Name



MAPPING REALITY TO PYTHON

How we translate real-world concepts into efficient Python classes:

Real World Context	Python Programming Concept
Student Name / Marks	Instance Variable (self.name)
Company Employee ID	Instance Variable (self.id)
School Name / Location	Class Variable (Static Context)
App Version / Tax Rate	Class Variable (Global to Class)

INSTANCE VARIABLES (PERSONAL)

Key Concept

These variables belong to **each object separately**. Changing data in one object does not affect any other objects created from the same class.

```
class Student:
    def __init__(self, name, marks):
        self.name = name # Instance
        self.marks = marks # Instance

s1 = Student("Saurav", 90)
s2 = Student("Rahul", 80)
print(s1.name) # Output: Saurav
```


CLASS VARIABLES (SHARED)

Key Concept

These variables belong to the **entire class**. They are defined outside the constructor and shared by all instances, saving memory and ensuring consistency.

```
class Student:
    school_name = "ABC School" # Class Var

    def __init__(self, name):
        self.name = name

s1 = Student("Saurav")
s2 = Student("Rahul")
print(s1.school_name) # ABC School
```


COMBINED IMPLEMENTATION

- ✓ **Shared Context:** The school variable is defined at the class level.
- ✓ **Specific Context:** name and marks are defined inside `__init__`.
- ✓ **Access Pattern:** All objects (s1, s2) can access the class variable using dot notation.

```
class Student:
    school = "ABC School"

    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

s1 = Student("Saurav", 90)
s2 = Student("Rahul", 85)
print(s1.name, s1.school)
```


GLOBAL STATE MODIFICATION

1

UPDATE AFFECTS
ALL

Changing Class Variables

If the school changes its name, we update the Class directly, not individual students:

```
Student.school = "XYZ School"
```

Now, `print(s1.school)` and `print(s2.school)` will both output **XYZ School**.
This acts like a company-wide rule change.

DEVELOPER INSIGHTS: BEST PRACTICES



Use Class Var For:

Global settings like **Game Difficulty**, **Tax Rates**, **App Version**, or **Company Name**.



Use Instance Var For:

Personal attributes like **User Balance**, **Individual Scores**, or **Cart Items**.



Memory Tip

Class variables are stored only **once** in memory, making them highly efficient for massive data sets.

THINK BEFORE RUNNING

Challenge: Predict the output.

```
class Car:
    wheels = 4 # Class Variable

    def __init__(self, brand):
        self.brand = brand

c1 = Car("BMW")
c2 = Car("Tesla")

print(c1.wheels)
print(c2.wheels)
```

Why do they both show '4'? Because wheels are a standard property of the Car blueprint!



DAY 8: PRACTICE TASKS

System	Class Name	Instance Vars (Personal)	Class Var (Shared)
School System	Student	name, marks	school_name
Company System	Employee	name, salary	company_name
Game Core	Player	name, health	game_name
Auto Industry	Car	brand, price	wheels = 4

Summary & Wrap-up

You've mastered the core of Python Object Modeling:

- ✓ Instance vs Class Variables
- ✓ Shared vs Personal Data Contexts
- ✓ Memory Optimization & Real-world Logic

`</>` Next Session: Methods & Decoration
Stay Curious, Keep Coding.

IMAGE SOURCES



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Source: learningmole.com



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